Description Editorial Solutions Submissions

704. Binary Search

Solved

Easy Topics

Given an array of integers `nums` which is sorted in ascending order, and an integer `target`, write a function to search `target` in `nums`. If `target` exists, then return its index. Otherwise, return `-1`.

You must write an algorithm with `O(log n)` runtime complexity.

Example 1:

> **Input:** nums = [-1,0,3,5,9,12], target = 9
> **Output:** 4
> **Explanation:** 9 exists in nums and its index is 4

Example 2:

> **Input:** nums = [-1,0,3,5,9,12], target = 2
> **Output:** -1
> **Explanation:** 2 does not exist in nums so return -1

Constraints:

- $1 <= \text{nums.length} <= 10^4$
- $-10^4 < \text{nums[i], target} < 10^4$
- All the integers in `nums` are **unique**.

Code

Python3 Auto

```python
from typing import List

class Solution:
    def search(self, nums: List[int], target:
        low, high = 0, len(nums) - 1

        while low <= high:
            mid = low + (high - low) // 2

            if nums[mid] == target:
                return mid
            elif nums[mid] < target:
                low = mid + 1
            else:
                high = mid - 1

        return -1
```

Restored from local Upgrade to Cloud Saving

Testcase Test Result

You must